

Design Proposal for MIME Capstone Design
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Team # MIME - 5504
NASA Psyche Space Elevator – Attachment

Psyche Asteroid – Electrostatic Anchor for Space Elevator Tether

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Introduction

Within astronomy and space exploration disciplines, mapping and collecting samples from asteroids is a complex and arduous task. This generally consists of launching multi-million-dollar space vehicles into the dark empty void to eventually make high precision rendezvous with extraterrestrial bodies. Once in situ an extremely risky landing sequence is initiated and is entirely dependent on the governing dynamic environmental mechanics of the celestial object. The likelihood of failure is astronomical.

So, to reduce the risk presented to space orbiter and lander vehicles, a space elevator has been proposed. Though a lofty concept, a space elevator poses significant advantages over traditional approaches to landing on extraterrestrial surfaces, including:

- Limiting risk exposure to the main space orbiter.
- Reduced landing costs.
 - Less chemical fuel.
 - Less expensive landing vehicles.
- Ability to stop lander descent at any altitude with limited power requirements.
- Providing a foundational framework for repeatable return visits.
 - Increasing applications for mining throughout and sample extraction.

These advantages compound, increasing its attractiveness to be a potential system for mining asteroids. However, before its implantation is possible, a series of major aspects need to be accounted for.

What is a space elevator? It is a cabled structure tens to hundreds of miles long and is anchored at opposing ends. One end is anchored to the surface of a celestial object like Earth (Terra), or for our design purposes, Psyche. The other end is connected to a space orbiter or station in a geostationary orbit. Many creative design concepts have been posed, most showing large oceanic cement and steel anchor plat forms and large satellite shipping hubs as seen in **Figure 1**. Yet, no real products have been implemented and the materials necessary for its real-world application are currently unavailable in the required amount due to manufacturing limitations. However, this is fast changing, and a lot of work has been done in the field of material sciences that are making strides towards developing the methods necessary to manufacture these space age materials. There is much research worldwide into the properties of carbon nanotubes (CNTs), graphene and other materials based upon carbon [1]. These types of materials are necessary for the development of the space tether.

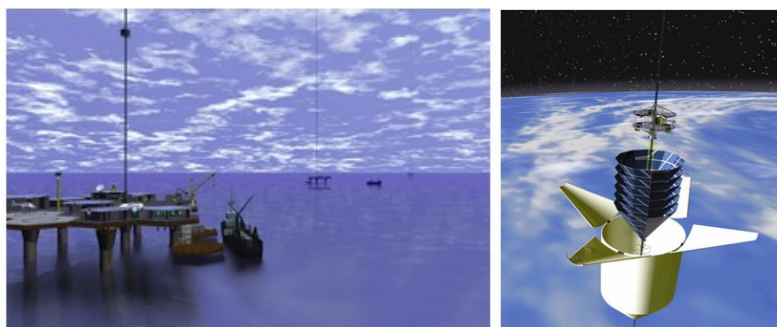


Figure 1. Space elevator foundations, and orbiting shipping hub

Project Scope

The goal of this project is two-fold:

- Broad Scope: To investigate the feasibility and utilization of space elevator technology for the purpose of mining Psyche.
- Refined Scope: To attach the base end of a space elevator tether to an anchoring apparatus that interfaces with the assumed solid ferritic surface of Psyche.

Psyche: What it is and Why it Matters

We first must become familiar with the interface we are proposing to attach a space elevator tether to. From Arizona State Universities (ASU) Psyche Mission FAQ's [2]:

- Psyche is an asteroid approximately 150 miles in diameter.
- Psyche is theorized to be 30-60% ferrous metal with more aspiring estimates suggest it is solid metal and the remnant core of an early planetesimal.
- Psyche is estimated to have an average surface gravity of 0.144 meters per second squared (m/s^2).
- Psyche is extremely dense with estimates ranging from 3400 to 4100 kilograms per meter cube (kg/m^3).
- Psyche's gravity appears to be about 1.5% of earth's gravity.
- Psyche experiences one solar day or one full rotation approximately every four hours.
- Psyche's high rotation rate implies an expected high thermal cycling rate with temperatures ranging from 200K to 75K between the day and night sides.

By understanding the in-situ surface interface and the dynamic load system we intend to operate in, we can glean useful information which will inform the design of the anchor. Such as:

- Materials – due to Psyche's high rate of thermal cycling we need materials that have relatively stable thermal expansion properties in this range.
- Kickback – due to Psyche's low gravity we have to consider how force loads applied to the surface will affect the landing vehicle's ability to stay in contact and not kick off pushing it back into space.

Psyche is a unique and important environment not only for its potential in scientific discovery, but for its abundance of metal resources. Unvalidated estimates place Psyche's value at ten quintillion dollars. Though the estimated value of Psyche sounds ridiculously high, its gravitas lends to Psyche's viability as a planetary resource. Alone, the remnant planetesimal could provide precious metals to sustain the world's economical requirements for the next millennia. It could also provide the means for jump-starting a large-scale space manufacturing industry. Sending materials into space is expensive, risky and ultimately exhaustive. Therefore, high density, low gravity, mostly metal behemoths like Psyche are primary candidates for humanity's growth into the stars. By utilizing Space elevator technology in conjunction with Psyche-like asteroids, we can develop a low-cost steady supply chain of ethically sourced metal resources without depleting Terran-based reserves.

Design Process/Approach to Problem

To reiterate, the problem we are presented with is:

- Attaching the base end of a space elevator tether to an anchoring apparatus that interfaces with the solid ferritic surface of Psyche.

To solve this problem, our design team utilized the following design process:

- First, perform preliminary research.
 - Determine governing environmental factors.
 - Analyze previous solutions to similar design problems.
- Second, gather, organize and categorize customer requirements.
- Third, turn customer requirements into engineering requirements.
- Fourth, develop solution concepts using a Morphology matrix.
- Fifth, analyze concepts utilizing a Pugh chart.
 - Statistically weighing concepts feasibility to design parameters.
 - Down selects concepts to inform design iteration.
- Sixth, prototype down selected concepts.
- Seventh, perform refining research based on prototypes successes and pitfalls.
- Eighth, iterate (repeat steps as needed).

In the preliminary design iteration, we scoped out various methods for attaching objects, including harpoons, nets, pinchers, wedges and drills using a morphology matrix seen in **Figure 2**. However, each requires physical connections and can have an array of mechanical problems.

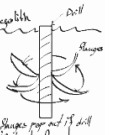
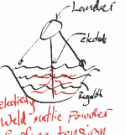
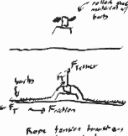
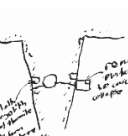
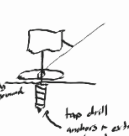
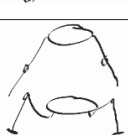
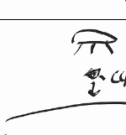
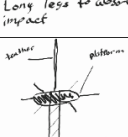
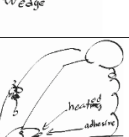
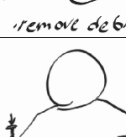
MORPH-MATRIX								
Project:		Psyche Space Elevator Attachment						
	Concept	Discriptions	Concept	Discriptions	Concept	Discriptions	Concept	Discriptions
Asteroid Surface Attachment Method		During landing the feet of the lander drill in using the inertia of the lander. Potential applications with the flanged drill head concept and self locking suspension		Self Tapping Metal Rod launched from orbit to impact into regolith with an anchor loops at the end to attach bottom of tether. Potential applications with flange drill head, Wedge and Platform Concepts		Drill has counter-rotational flanges that extend out to dig into the material locking it into place as upward force is applied and/or the drill is backed out		Lander uses an electrode forced into the mostly metal regolith using a supercapacitor connected to the electrode will discharge electrical potential to heat up and melt the metal, welding the lander to the surface. Potential use of Flange drill to increase the surface area of the electrode
		During landing, sheets of material are shot outwards from the lander. The goal is to maximize surface area contact with the surface of Psyche to create a large amount of friction. This friction is further increased by barbs and material choices. Hopefully, the rope like/elastic nature of the material causes tension to be diverted to convert the upward pull of the tether into a horizontal pull that can be countered by friction.		Remove material from an existing steep crater to cause a collapse that holds the anchor and craft down. Alternatively, the regolith and/or core metal could be melted above the crafts legs to hold it in place. This melting could be done with thermite or other welding methods.		This is a spin on a drill based approach, where the conventional drill bit is replaced with a special bit that is similar to a screw tap. This would mean that the drill would pull itself deeper upon initially entering the material, and there is potential for it to remove material to the surface by spinning in the opposite direction (in the same way that a screw tap turns 4 clockwise and 1 counter clockwise).		Upon landing, the landing leg shoots out a metallic ballistic designed to penetrate Psyche's regolith and reach the metallic core. There would be a strong tether/cha in attaching the ballistic to the lander. Upon hitting the metallic core of Psyche, a thermite payload would be ignited to weld the metallic ballistic to the Psyche's metal core.
		As we land, the legs will have suspension springs inside that will be self locking as they move down. The legs lock in order to remove any of the kinetic energy of the springs to launch the spacecraft back upwards		The legs will have two large segments with a hinge that will absorb the force of the impact. Since the gravity is ~1% Earth's gravity the total force experienced should be small.		Using the right surface location, we can design our spacecraft to wedge into a crater or canyon. the sides of the crater could also have loose debris that we could use to 'bury' the landing gear slightly to hold the spacecraft to the surface		By using an explosive device on the surface of Psyche could remove unnecessary debris that could make it impossible for us to drill/weld/etc into the surface to maintain surface connection. This debris that is blown can also be netted up to gather small samples of surface material. The man made crater could also be used to as a safer landing area.
		Two-stage attachment probe that first magnetically attaches to asteroid surface with three legs, then has a second stage of three legs that have heated suction cups attach to the asteroid and spray adhesive to attach. The probe would then drill down from the center of it below into the asteroid		A large spinning drill launched from the orbiter down into the asteroid, with a large platform deploying and stopping the drill once it makes contact with the surface. The tether will stay attached to the platform		The three legs of the probe all have an adhesive where they make contact with the asteroid, then heating to such a point that it melts into the iron. A stake is then attached to complete the anchor		Each foot of the anchor will actively drill into the asteroid, with a stake inserted from above into the drilled holes

Figure 2. First iteration design morphology matrix for concept generation.

One innovative device that shined over all the others is the Electro-Static gripper (ESG) **Figure 3**. ESG's have been used in the commercial Semiconductor Manufacturing and in package handling [3] industries and have recently begun to make appearances in space attachment approaches for its flexibility and versatility. ESG's also come in various shapes, sizes, and orientations to maximize gripping forces. Electroadhesion can enable lightweight, ultra-low-power, compliant attachment in vacuum by using an electrostatic force to adhere to similar and dissimilar materials [3].

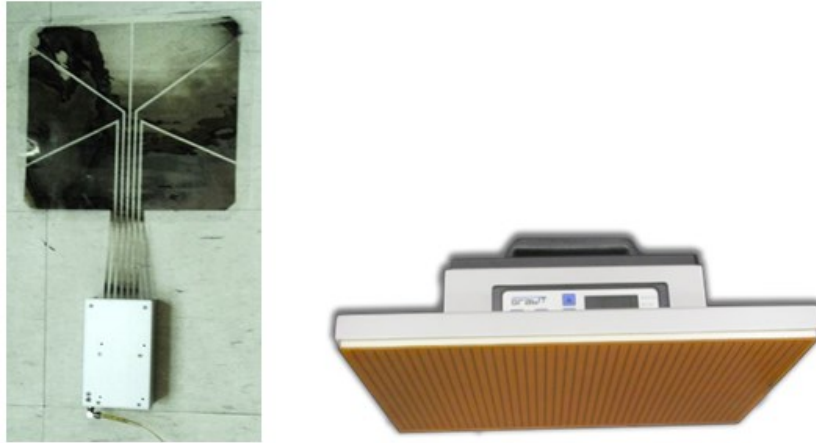


Figure 3. Industrial ES/EA Grippers From ElectroGrip, Co. and GrabIt, Inc.

Another method that shined in the concept generation phase is welding the anchoring device to the surface of Psyche. Though there are several methods for welding including:

- Cold-welding
- Ultra sonic
- Friction
- Induction
- Controlled explosive

Controlled explosive welding **Figure 4**. is the most promising welding method for our application. Explosion Welding is a solid-phase welding process in which explosives are used to combine components using high energy expansion waves to force components to fuse. With over 260 various combinations of similar and dissimilar metals and alloys having been successfully fused using this process [4].

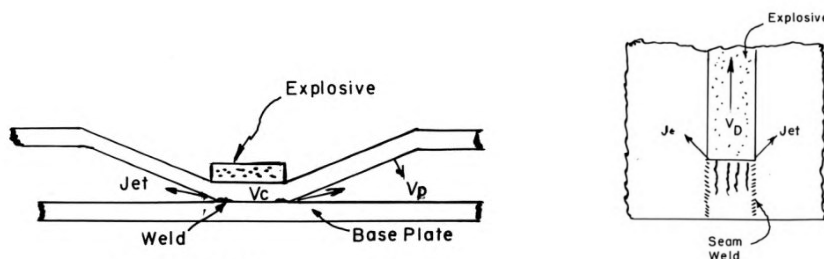


Figure 4. Schematic sketch of parallel plate configuration for NASA-LaRC explosion seam welding.

Both the ESG and explosive welding concepts offer our team a holistic and creative approach to move towards in proceeding design iterations. This is because utilizing ESG's as a subsystem in the anchoring device allows the system to have soft-point docking applications for the tether to surface interface. Implying that the anchoring device can be adjusted in-situ to optimize desired anchoring or mining locations. While the explosive welding subsystems located throughout the structural framework would allow for hardpoint docking, making the anchor a permanent fixture on the surface.

Sixteen unique concepts were generated in a morphology matrix. They were then statistically analyzed using a Pugh chart as seen in **Figure 4**. Regarding the following design criterion formulated from customer requirements:

- Lightweight.
- Cheap to manufacture.
- Ability to anchor with uneven surfaces.
- Ability to anchor to solid ferritic surfaces.
- Concept feasibility.
- Compatibility with space tether climber and sample delivery subsystem.
- Attachment speed.
- Successful attachment probability.

The ESG's and explosive welding were analyzed and given a weight average score greater than 80% based on their compatibility with the above design criterion. This provides the design with a solid direction in which to start prototyping.

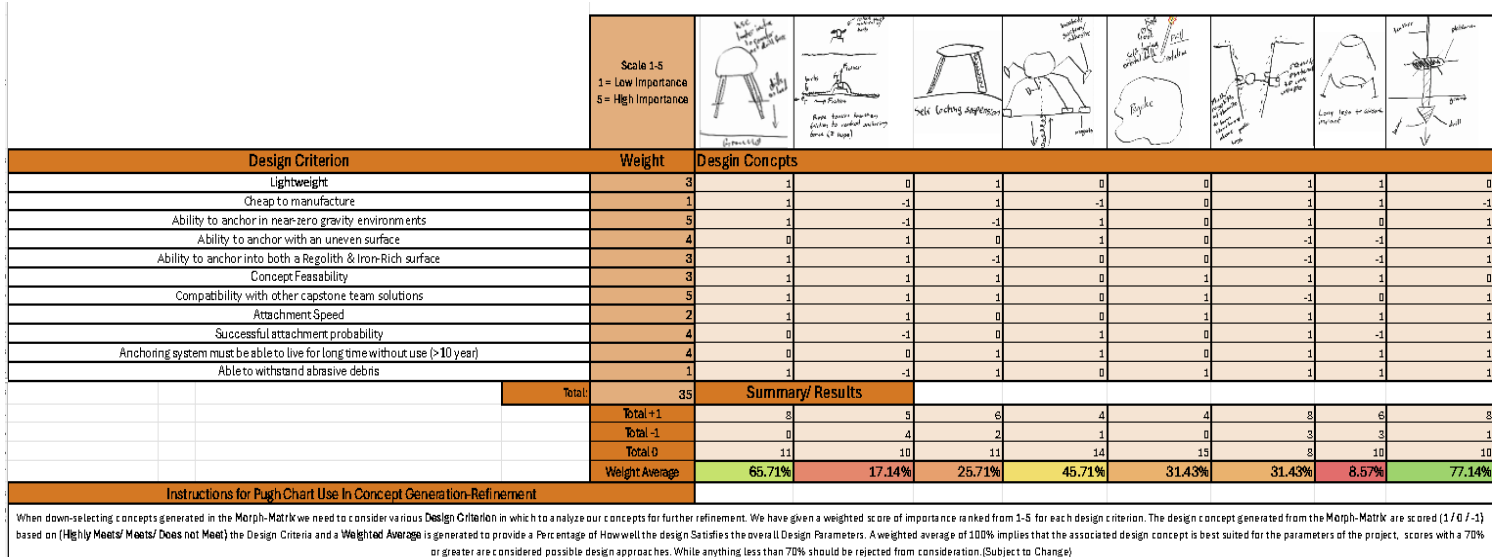


Figure 4. Shows partial PUGH chart for first design iteration used to down select concepts for compatibility to design criterion.

Proposed Design Solution

The anticipated design solution will utilize ESG's panels fixed to triangular living hinged mesh structures that will serve as the anchoring interface. The triangular living hinge mesh structures will allow for the panels to mold to most any irregular surface, optimizing the ESG's ability to make a proper connection point. The ESG that will be inspiring our future designs is shown in **Figure 5**. This ESG operates at 3 kV and has an electrostatic force of 3.5 N on Ge-coated polyimide film [5]. When considering its effectiveness on nonferrous surfaces, its potential anchoring force on ferritic surfaces is estimated to be greater. The triangular mesh structure will also allow it to fold up and away when not in use. At strategically located nodal points, controlled explosive welding devices will be integrated for hard docking applications. In **Figure 6**, is the low fidelity proof of concept highlighting the key aspects of the proposed design. The rectangular frame is representative of a black box which will house the power supply, motors, actuators and computer guided controls. The orange cloth shows the ability for living hinges to conform the feet structure to various orientation and surface conditions.

Figure 5. Fabricated electrostatic gripper and schematic highlighting electrode pattern.

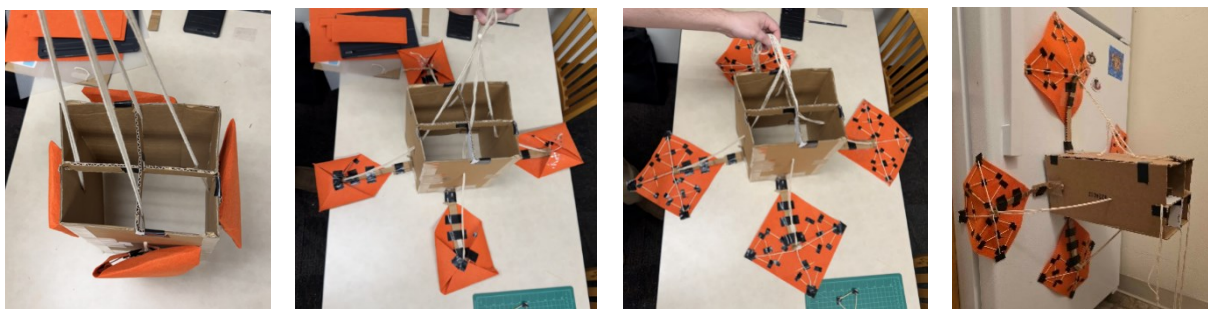
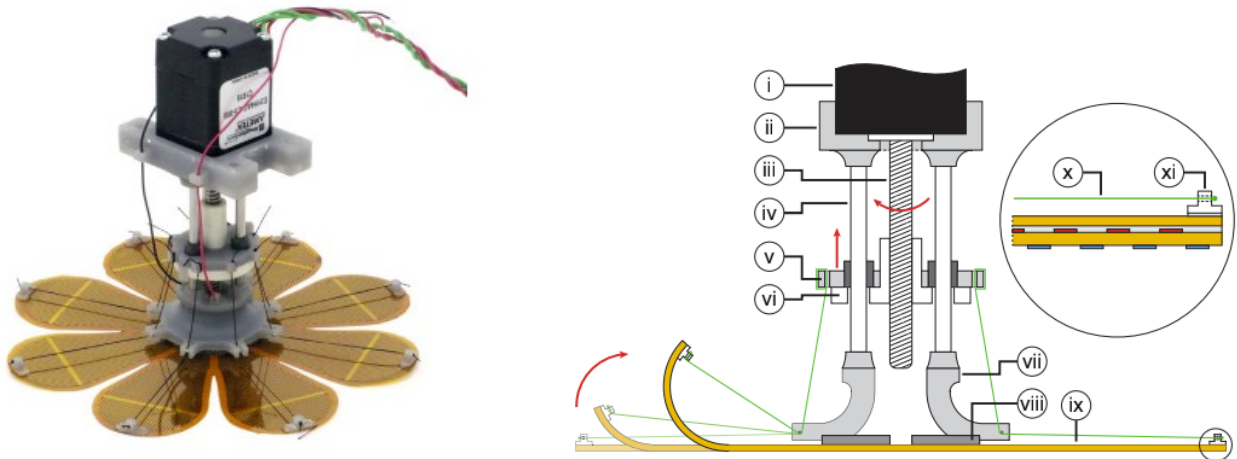


Figure 6. First iteration low fidelity prototype showing deployment process and mounting capability. Black tape represents nodal locations for explosive welding devices.

Secondary iterations of this design include a gimbed attachment between the tether and the landers main body shown in **Figure 7**, and a balled elbow joint in the legs to allow for the landers feet to individually adjust to surface conditions shown in **Figure 8**.

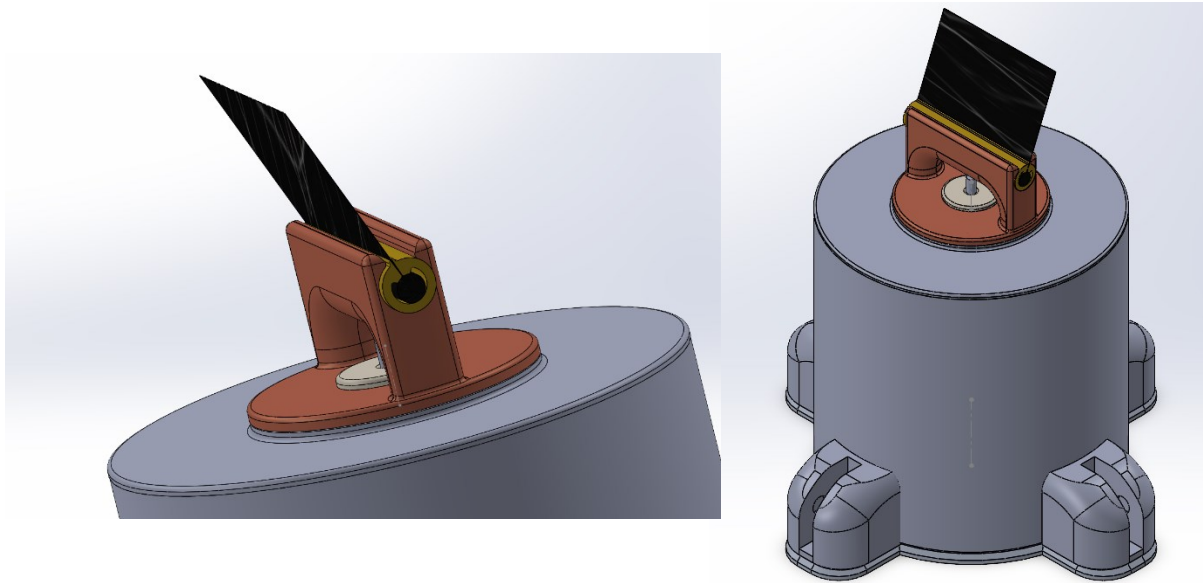


Figure 7. Tether gimbal attachment

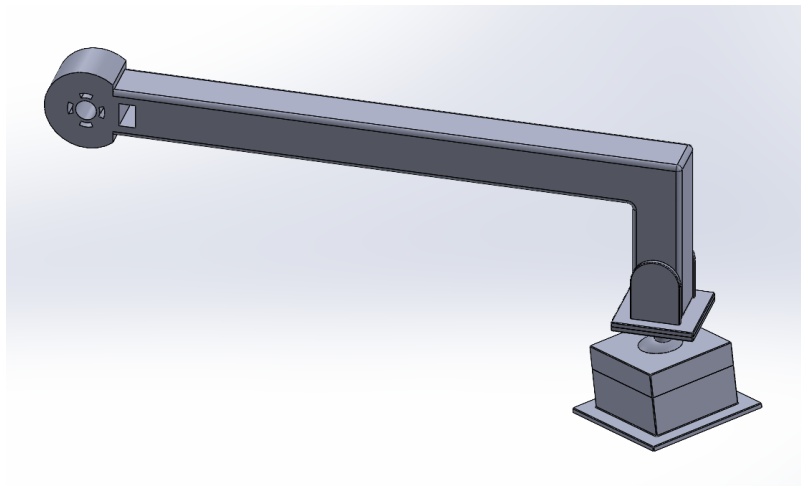


Figure 8. Second iteration leg design

This system has an array of potential benefits as well as risks that we must consider when iterating further:

- Pros – what are the benefits?
 - Attachment of tether has 360 degrees of axial rotation about the lander's central vertical axis.
 - Attachment of tether allows for 24 degrees of rotational pitch about the horizontal axis.
 - Ball joints allow for dynamic foot placement on uneven terrain.
 - Electrostatic Gripping feet allows for soft docking applications increasing the systems ability to redeploy if original surface conditions don't allow for proper attachment
 - Explosive welding allows for hard docking. Making the anchor a permeant fixture.
- Cons – what are the Risks?
 - Electrostatic grippers require kV of electricity to functionally operate. Which indicates large power requirements. However, this can be mitigated by running a low the electrical current through a series of transformers.

- Explosive welding can cause contamination of the surface, which can interfere with the integrity of the mineral samples collected in the surrounding vicinity of the lander. We can mitigate this by directing the exhaust of the explosive combustion up and into space or in the opposite direction of the intend sampling area.
- Leg positioning can lead to complications in landing because of the complexity of point-to-point contact between all four legs and the surface. It is necessary to actuate the legs independently.

Conclusion

Our design goal is to attach a space tether to the surface of the potentially iron dense planetesimal remanent, known as Psyche. The proposed design solution is to use a series of electrostatic gripping devices in conjunction with explosive welding between the anchoring structure and the metal surface of Psyche. Though there are several risks and a plethora of unknown variables, the design team has implemented proven engineering and design practices in order to develop conceptual design technologies into a potentially viable product design.

Work Cited

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